

Enterprise Analytics Architecture

- **Project Title:** AdventureWorks Enterprise Sales Analytics Platform
- **Document Version:** 1.0
- **Domain:** Architectural & Platform Engineering
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- **Date:** June 2026
- **Status:** Approved for Implementation

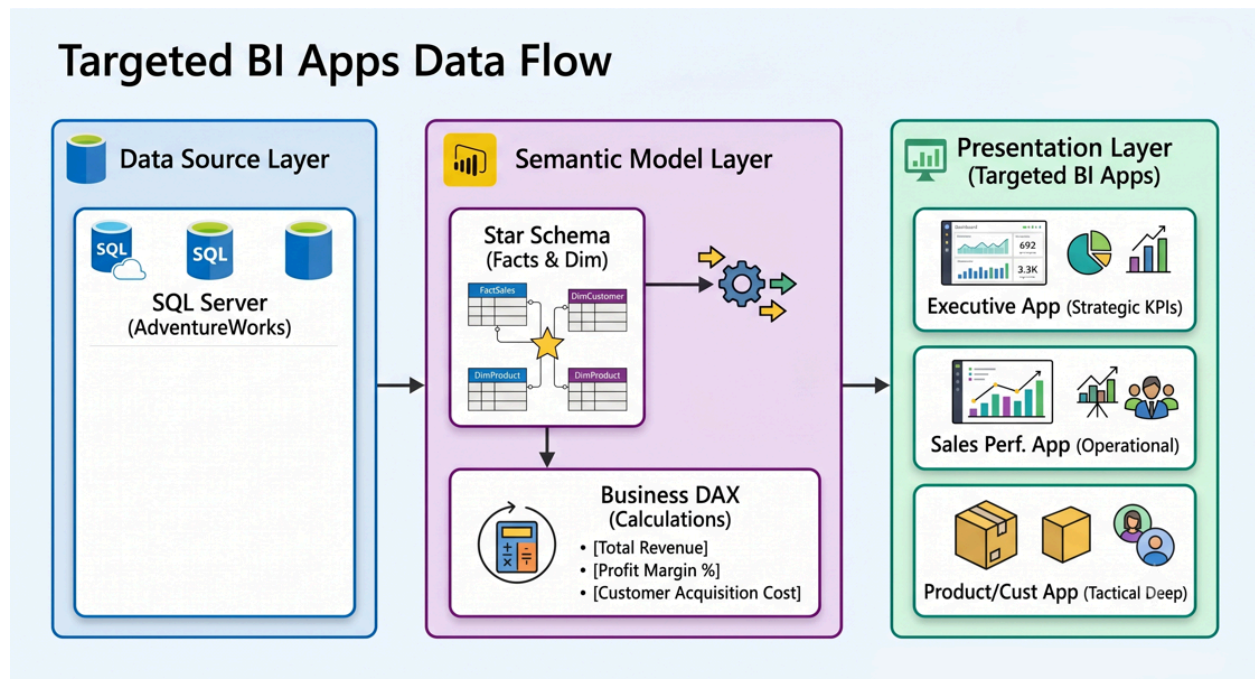
1. Architectural Executive Summary & Purpose

The purpose of this document is to outline the end-to-end data engineering and analytics architecture for the AdventureWorks Enterprise Sales Platform.

In enterprise environments, data model sustainability relies on a clear separation of concerns. This architecture defines how raw relational data from the source transaction systems is extracted, structured into a high-performance **Kimball Star Schema**, enriched with standardized business logic (DAX), and packaged into specialized analytical applications for diverse business stakeholders.

2. Enterprise Data Flow & System Topology

The platform coordinates the movement of data from database storage to business insights through a structured, unidirectional pipelines:



3. High-Level Architecture Diagram

4. Architectural Layers Detail

This architecture is organized into six highly cohesive, loosely coupled layers:

Layer 1: Data Source Layer (Storage & Ingestion Base)

- **Technology:** Microsoft SQL Server.
- **Component:** AdventureWorksDW (Enterprise Data Warehouse schema).
- **Characteristics:** * Features pre-optimized, cleaned historical and transactional records.
 - Serves as the immutable source of truth for downstream analytical assets.
 - No transformations or structural changes are made directly at this source level to ensure data lineage and database integrity.

Layer 2: Data Storage & Connection Layer (Logical Extraction)

- **Methodology:** Direct Extraction via **Live Connection Mode**.
- **Design Rationale:** * To bypass the need for a separate physical Data Lake in Version 1.0, data is ingested directly from SQL Server into the Power BI workspace.
 - **VertiPaq In-Memory Storage Engine:** Data is highly compressed and loaded directly into RAM, enabling sub-second visual rendering times and reducing the operational transactional query load on the primary SQL Server.

Layer 3: Data Modeling Layer (Semantic Star Schema)

- **Methodology:** Star Schema Dimensional Modeling (Kimball Design).
- **Structural Components:**
 - **Fact Tables:** Partitioned by transactional boundaries (FactInternetSales, FactResellerSales) and corporate budgets (FactSalesQuota).
 - **Conformed Dimensions:** Flattened tables (DimDate, DimProduct, DimCustomer, DimSalesTerritory) that establish a consistent vocabulary across all tables.

Layer 4: Business Logic Layer (Unified Calculations)

- **Technology:** DAX (Data Analysis Expressions).
- **Concept:** **Centralized Business Logic Assembly**.
- **Characteristics:**
 - Eliminates the risk of report-level custom calculations.
 - Standardizes complex formulas like Margin %, Quota Achievement %, and Time Intelligence (YoY / MoM growth) inside the model, making them automatically available to all downstream visuals.

Layer 5: Semantic Model Layer (Consolidated Power BI Dataset)

- **Component:** Unified Enterprise Sales Semantic Model (Metadata & Relationships).

- **Characteristics:**

- Maps physical relationships (strictly enforcing $1 : \infty$ cardinalities and single-direction filtering).
- Houses the model hierarchies (Category $\rightarrow$ Subcategory $\rightarrow$ Product Name) and metadata translations.
- Configured as a **single, shared cloud dataset** on Power BI Service, allowing multiple report creators to connect simultaneously without duplicate data loads.

Layer 6: Presentation Layer (Tailored BI Applications)

Instead of delivering a single, cluttered dashboard, the architecture splits consumption into three specialized, role-based Power BI Applications:

1. **Executive Performance App:** A highly summarized dashboard displaying high-level metrics (Revenue, Profit, YoY Margins) tailored for C-Level executives.
2. **Sales Operations & Performance App:** Tactical, drill-down workspace tracking sales quotas vs. actual achievements, built for Regional Sales Managers.
3. **Product & Customer Analytics App:** Deep-dive exploratory reporting highlighting purchase behaviors, customer Lifetime Value (LTV), and product velocity trends.

5. Core Architectural Design Principles

To safeguard data reliability and system performance, five core design principles are strictly applied:

- **Kimball Star Schema Design:** Relational schemas are flattened. Snowflaking is avoided to optimize the column-oriented VertiPaq database storage.
- **Single Directional Filtering:** Filters flow exclusively from Dimension tables (1) to Fact tables ($*$). This ensures predictable query behavior, prevents calculation loops, and lowers processing overhead.
- **Conformed Dimensional Integrity:** Shared dimensions use identical keys and attributes, ensuring that B2C and B2B revenues can be safely aggregated and evaluated side-by-side.
- **Centralized Business Logic (Single Source of Truth):** All metrics exist as global DAX measures within the semantic dataset. Under no circumstances are hard-coded calculations written directly into individual visuals or separate report sheets.
- **Business-First Modeling Approach:** Table and column names are translated from technical database labels (e.g., EnglishProductName) to user-friendly terminology (Product Name) and grouped into functional display folders to support intuitive self-service analytics.

6. Architectural Rationale: Why This Design?

Architectural Goal	Technical Decision	Business Value & Justification
High Performance	Import Mode + Star Schema	Guarantees immediate, interactive report rendering (<1 second) even when slicing millions of records across multiple dimensions.
Metric Consistency	Centralized DAX Semantic Layer	Eliminates reporting discrepancies. Financial, operational, and executive teams will always see the exact same margin and growth values.
Operational Efficiency	Separated Presentation Layer	Relieves pressure on SQL Server transaction engines. Resource-heavy calculations are handled in the BI cloud layer, not on operational servers.
Enhanced Security	Role-Based Reporting (Apps)	Segregates operational visibility. Executives see big-picture numbers, while regional managers focus exclusively on their team's data.

7. Target State Architecture & Upgrade Path

This Version 1.0 architecture is designed to scale horizontally. As AdventureWorks matures, the platform will migrate to a modern, unified lakehouse architecture:

- **Transition to Microsoft Fabric (OneLake SaaS):** Future phases will move data warehousing from standard relational servers to Fabric's Lakehouse. This allows the semantic model to run in **Direct Lake Mode**, reading Delta tables directly from OneLake storage in real-time without needing traditional Import refreshes.
- **Multi-Domain Data Product Convergence:** Integrate the upcoming **Inventory Analytics** and **Finance Analytics** domains as independent data products, joined seamlessly to the core model via the conformed dimensions.
- **Granular Row-Level Security (RLS):** Implement RLS models within the shared dataset to

restrict data visibility so that regional sales representatives can only view transactions occurring inside their designated sales territories.